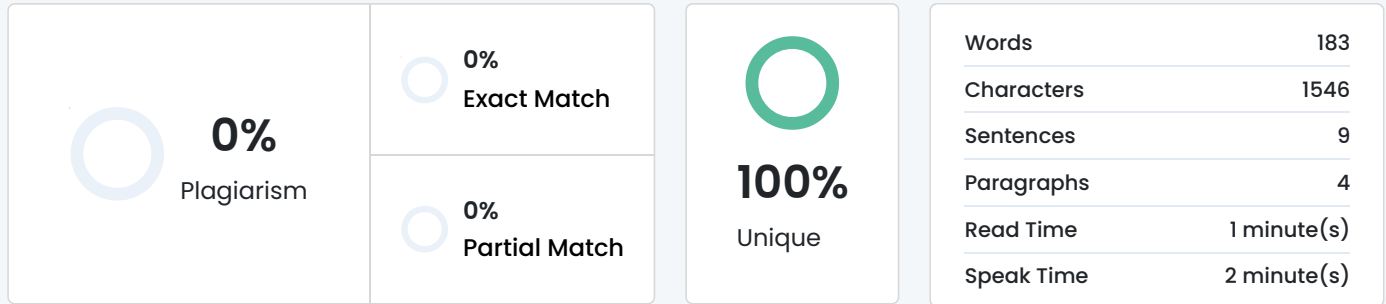


Plagiarism Scan Report



Content Checked For Plagiarism

Unlike many existing gesture-to-speech systems that rely on flex sensors or image processing techniques, the proposed Gesture-to-Voice Converter introduces a combinational contact-based sensing mechanism for detecting finger gestures. Instead of measuring analog bending values through expensive flexible sensors, this system uses a structured digital contact circuit that detects specific finger combinations through controlled electrical connections. This design significantly reduces system complexity and overall cost while maintaining reliable gesture recognition accuracy.

Most previously developed systems depend on camera-based recognition or machine learning algorithms, which require high processing power and advanced hardware. In contrast, the implemented solution follows a deterministic logic-based mapping approach, where predefined digital patterns correspond directly to stored phrases. This ensures faster execution, lower latency, and improved stability in real-time operation.

Furthermore, the system integrates motion-based gesture detection using the MPU6050 sensor to expand communication capabilities beyond static finger combinations. The dual-input mechanism—combinational finger detection and motion orientation sensing—enhances flexibility without increasing hardware dependency.

By combining low-cost hardware, structured embedded programming, and wireless communication through Bluetooth, the project demonstrates a practical and scalable assistive communication model suitable for educational and social deployment.

Matched Source

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